VISHWAKARMA AWARDS
A Maker Bhavan Foundation Initiative

MAKER BHAVAN FOUNDATION

VISHWAKARMA AWARDS 2026-27 · STAGE 2

AI-Powered Environmental Monitoring Network

Solar-powered LoRa edge-AI nodes for early warning of floods, forest fires and air pollution

Vishwakarma Application ID	Theme	Sub- Theme
VK26-IITI-AHM-0602	AI in Hardware &Manufacturing	Edge AI & On-Chip Processing

Problem Context: Overview & Statement

Project Overview

Title	AI-Powered Environmental Monitoring Network
Application ID	**VK26-IITI-AHM-0602**
Track	AI in Hardware & Manufacturing
Core innovation	One low-cost solar node that fuses flood, fire-precursor and air-quality sensing, decides on-device with TinyML, and alerts over long-range LoRa.

Problem Statement & Target Beneficiaries

- **Pain point:** floods, forest fires and PM2.5 spikes are detected late or not at all in rural, forest-fringe and floodplain areas. Reference air stations are urban and cost about ₹1 crore each; satellite fire alerts depend on overpass timing and cloud or smoke cover; remote river stretches have no local gauge.
- **Beneficiaries:** forest-fringe and floodplain villages (gram panchayats), State and District Disaster Management Authorities, forest divisions and fire watchers, and local health and pollution-control offices.
- **Gap to close:** an affordable, ground-truth, multi-hazard node that warns within minutes where no infrastructure exists.

Scope: one pilot cluster (up to 10 nodes + 1 gateway) planned for Stage 3 field validation at a forest-fringe or floodplain site (to be confirmed). All performance values are design-stage estimates.

Problem Context: National Scale & Gap

PROBLEM CONTEXT

40 Mha

flood-prone land (about 12% of India). Average 1,600 lives and ₹1,805 crore lost per year (NDMA)

27.97%

of forest cover is highly to extremely fire-prone (ISFR 2023)

34,562 km²

of forest burnt in the Nov 2023–Jun 2024 fire season (FSI)

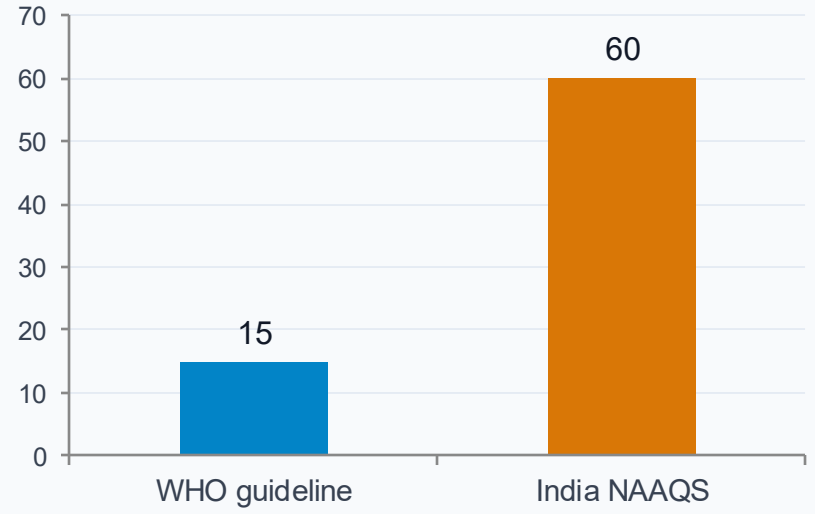
33,000+/yr

deaths linked to PM2.5 above the WHO guideline in 10 cities (Lancet Planetary Health, 2024)

Why current monitoring leaves a gap

- **Satellite (FSI SNPP-VIIRS alerts):** wide coverage, but cannot see river stage or ground-level PM2.5, and depends on overpass and cloud or smoke cover.
- **Reference air stations:** accurate but about ₹1 crore each plus ~10% yearly upkeep; only 261 continuous stations nationwide (roughly one per 12,600 km²).
- **Manual watch and sparse gauges:** remote forest blocks and upstream catchments stay unobserved.
- **Opportunity:** dense, ~₹7k nodes with 80–90% agreement to reference monitors after calibration (Eos, 2022).

PM2.5 24-hour limit (µg/m³)



Sources: NDMA flood guidelines; FSI/ISFR 2023 via MoEFCC and IndiaSpend; Lancet Planetary Health 2024; Eos 2022; The Wire Science. See slide 20.

User Journey: Before State

USER JOURNEY · BEFORE

FLOOD	Heavy rain upstream; no local river gauge near the village	People notice the river rising by sight or phone calls	Warning, if any, comes from a distant station or word of mouth	Evacuation begins after inundation: crops, livestock and lives at risk
FOREST FIRE	Ignition in a forest block (about 9 in 10 fires are human-caused)	Burns unseen until smoke is spotted or a satellite passes	Alert reaches the forest division, then a crew is mobilised	Fire spreads across blocks: 34,562 km² burnt in 2023–24 season
POLLUTION	PM2.5 or gas spike in a town or industrial belt	Nearest reference station is km away, or there is none	No hyper-local reading and no public alert	Unnoticed exposure: 33,000+ deaths per year in 10 cities

Quantified inefficiency

₹1 crore
per reference air station, plus ~10% yearly upkeep

1 per 12,600 km²
continuous air stations (261 nationwide)

60 vs 15 µg/m³
India NAAQS vs WHO PM2.5 24-h limit

1,600 lives
average annual flood toll, plus ₹1,805 crore damage

Flow steps summarise the national situation described in slide 3; sources on slide 20.

User Journey: After State

USER JOURNEY · AFTER

FLOOD	JSN-SR04T reads water level every 5 minutes	Edge rule: rapid rise switches the node to 30-second alert mode	LoRa alert packet reaches the gateway, then SMS and dashboard	Village and DDMA warned while the water is still rising
FOREST FIRE	Temperature rise, humidity drop and PM2.5 or gas spike detected	TinyML plus fusion confirms; no single-sensor trigger	GPS-tagged alert sent over LoRa and gateway GSM	Forest guard dispatched to a located node, early
POLLUTION	PM2.5 and MQ-135 gas index sampled on schedule	Readings mapped to CPCB AQI bands	SMS and dashboard alert when PM2.5 exceeds 90 µg/m³ (AQI “Poor”)	Local advisory for schools and residents

Efficiency (design targets)

5 min → 30 s
sampling interval, normal to alert mode

0.37 s
LoRa airtime for a 24-byte alert at SF10 (calculated)

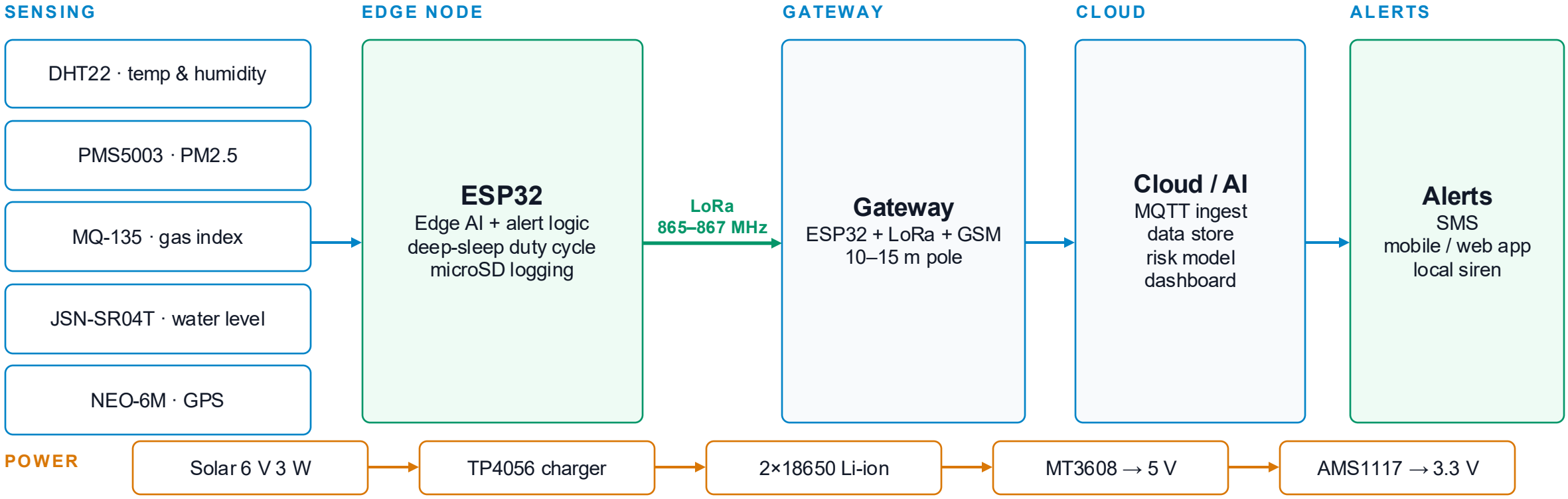
< 2 min
target detection-to-SMS time; to be verified in Stage 3

₹6,820 per node
about 1/1,500 of a reference station's cost

Low-cost sensors are not reference grade: PM2.5 and gas readings need collocated calibration before they are used for compliance decisions.

System Architecture: Block Diagram

SYSTEM ARCHITECTURE



Mechanical structure

IP65 ABS enclosure, radiation shield, 2 m GI mast, sensor boom

Actuation / power

Solar charging, Li-ion pack, power-gated sensors; siren output at gateway

Electronic controllers

ESP32 node controller; ESP32 + SIM GSM gateway

Firmware / logic flow

Wake → read → edge-AI check → alert (LoRa) or sleep

System Architecture: Target Specifications

Parameter	Target spec	Selected hardware / protocol
Controller	240 MHz dual-core; deep-sleep mode	ESP32-WROOM-32
Long-range link	865–867 MHz; est. 2–5 km open, 0.5–1 km in forest; sensitivity −132 dBm at SF10	RFM95W (SX1276) LoRa + gateway
Temp & humidity	±0.5 °C typ.; RH ±2–5 % (datasheet)	DHT22
PM2.5	0–500 µg/m³, laser scattering (datasheet)	PMS5003
Gas index	Relative air-quality index; needs calibration	MQ-135
Flood sensing	25–450 cm range, IP67 probe; speed-of-sound corrected using DHT22	JSN-SR04T ultrasonic
Location	GPS tag for each node; ~2.5 m CEP (datasheet)	NEO-6M
Sampling	Every 5 min; 30 s in alert mode	RTC timer wake-up
Battery backup	~4.5 days est. with duty cycling, before solar	2× 18650 (5200 mAh) + 6 V 3 W solar
Edge AI	Int8 model ≤ 50 KB, inference < 50 ms (targets)	TinyML on ESP32
Enclosure	IP65; radiation shield; 2 m mast	ABS enclosure + GI mast

Runtime estimate: avg. current ≈ (450 mA × 30 s + 1 mA × 270 s) / 300 s ≈ 46 mA, so 5200 mAh ÷ 46 mA ≈ 113 h (quoted conservatively as ~4.5 days). Solar: 6 V 3 W × 4.5 peak-sun-h × 0.7 ≈ 1.6 Ah/day vs load 1.1 Ah/day, a 1.4× margin (design estimate).

Hardware Selection & Protocols

Interface map and integration notes

Module	Interface	Integration note
DHT22	Single-wire GPIO	10 kΩ pull-up; ≥ 2 s read gap
PMS5003	UART2, 9600 baud	5 V supply; SET pin sleeps fan and laser
MQ-135	Analog, ADC1	Divider to ≤ 3.3 V; heater power-gated
JSN-SR04T	TRIG / ECHO GPIO	5 V echo divided to 3.3 V
NEO-6M	UART1 (remapped)	9600 baud; hot start for fast fix
RFM95W	SPI (VSPI)	3.3 V; 865–867 MHz channel plan
microSD	SPI, shared bus	Own chip-select; buffers unsent alerts

Analog inputs use ADC1 only (ADC2 is unavailable while Wi-Fi is active, e.g. during firmware updates); strapping pins 0, 2, 12 and 15 are left free.

Why LoRa for the uplink

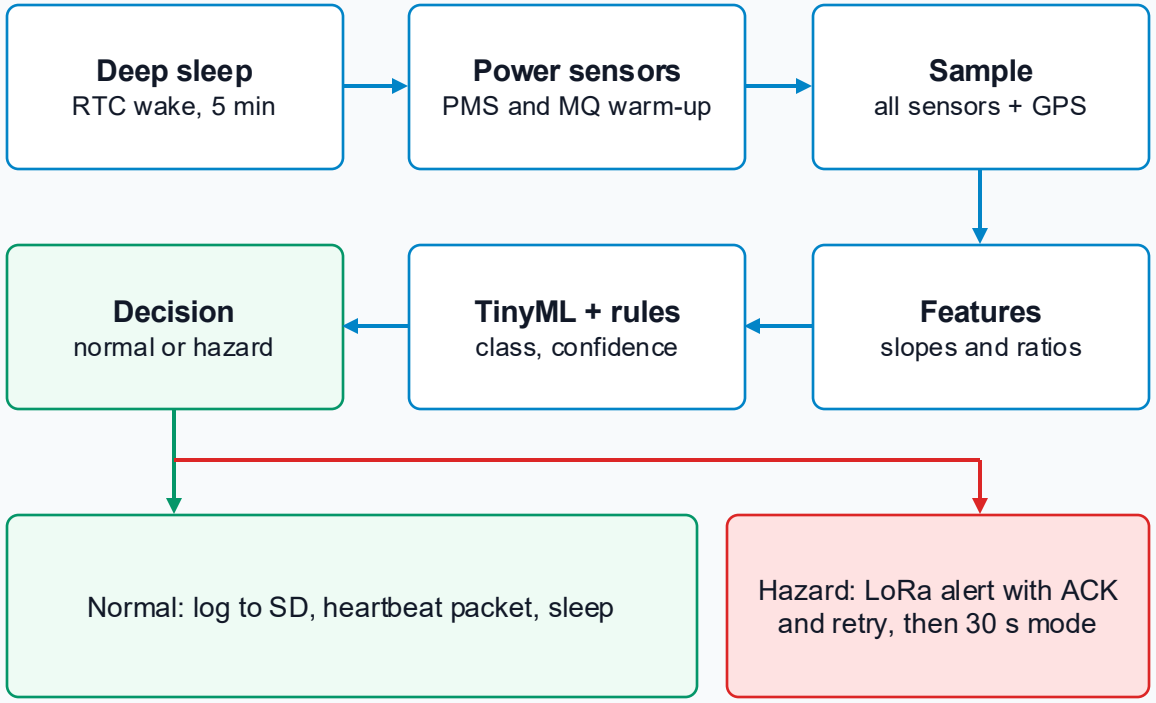
Link	Range	Power	Needs
LoRa	km-scale	Very low	1 gateway
Wi-Fi	50–100 m	High	Access point
GSM / NB-IoT	Coverage-dependent	Medium–high	Cell signal
Zigbee / BLE mesh	10–100 m per hop	Low	Many relays

- India permits 865–867 MHz LoRa at up to 1 W transmit power, 4 W ERP and 200 kHz carriers; the node transmits at 17 dBm (50 mW).
- ESP32 gives dual-core TinyML headroom, deep sleep, three UARTs and SPI at about ₹600.

Architecture rationale: LoRa chosen over Wi-Fi/GSM because forest cellular coverage is unreliable; only the gateway needs GSM or internet. ESP32 chosen for TinyML support and deep-sleep mode.

Edge AI & Firmware Logic Flow

Duty-cycle state machine



Hazard logic (initial rules, tuned on field data)

Hazard	Inputs	Initial trigger
Flood	Water level slope	Rise above site-set cm per 10 min, or danger mark crossed
Fire	ΔT, ΔRH, PM2.5, gas	Temp rise ≥ 5 °C in 10 min and RH drop, plus PM2.5 or gas spike
Pollution	PM2.5	> 90 µg/m³ on 3 consecutive samples

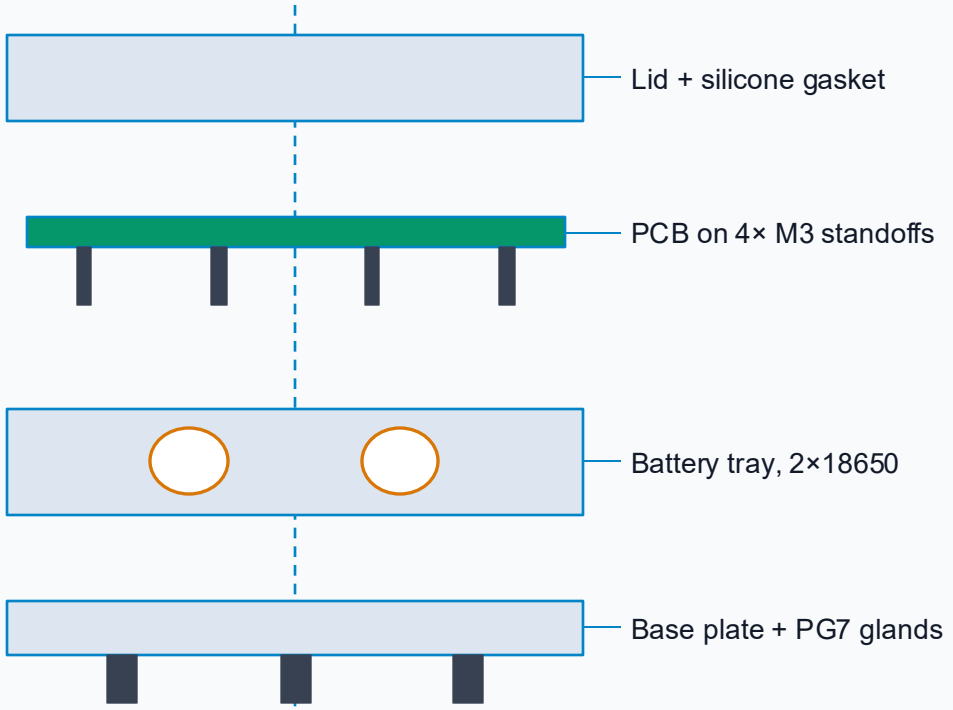
Alert packet, 24 bytes: node ID (2) · time (4) · lat/lon (8) · temp (2) · RH (1) · PM2.5 (2) · gas (2) · level (2) · flags (1). The classifier adds a class and confidence to the flags.

Validation targets (not yet achieved): recall ≥ 90% on fire and flood events and ≤ 1 false alarm per node per month, measured on bench and pilot data in Stage 3. Trigger values above are engineering assumptions.

CAD Design: Concept Views

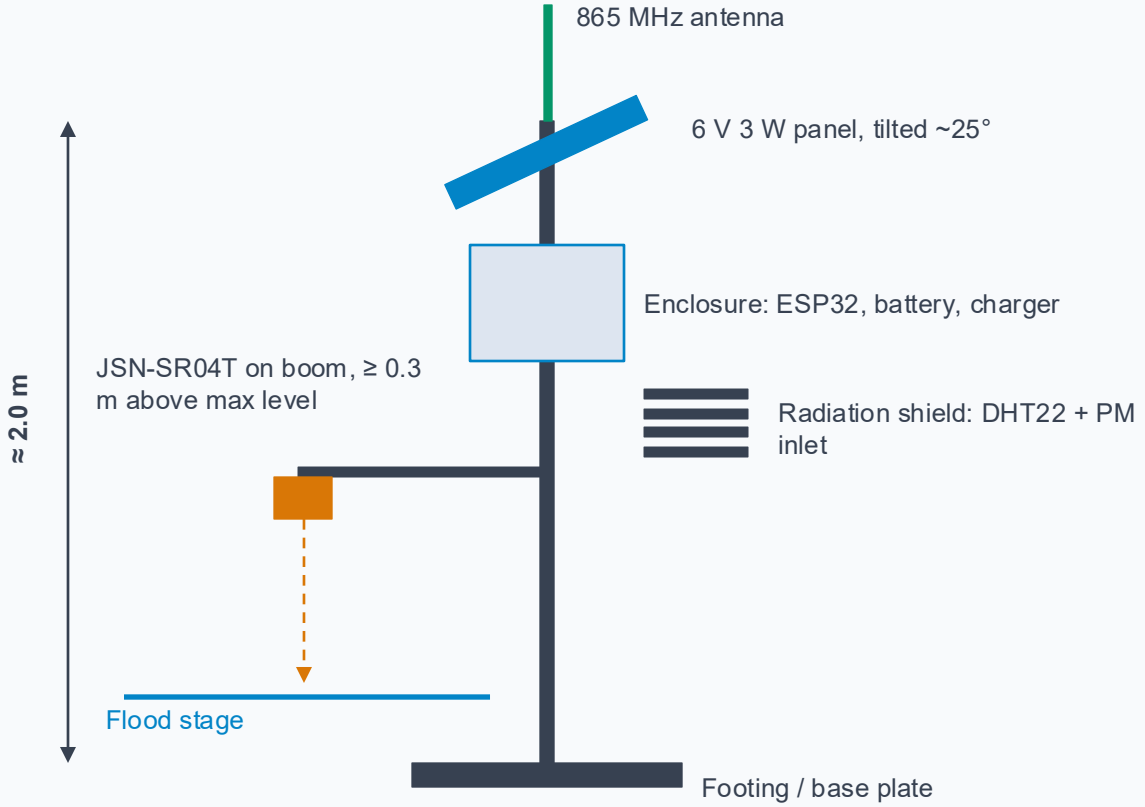
CAD & MECHANICAL

Enclosure exploded view (concept)



Typical IP65 junction box, about 160 × 120 × 70 mm; sensor cables exit through gasketed PG7 glands.

Mast assembly, side elevation (concept)



Concept-level views drawn for this submission. Parametric native models (.STEP / .SLDPRT) belong in the Maker Journey repository (slide 15).

Mechanical Specs: Materials & Structural Check

Part	Options considered	Selected	Why
Enclosure	ABS, ASA, polycarbonate	ABS, IP65 (ASA if UV-limited)	Low cost, easy to drill, gasketed lid
Brackets, mounts	PETG, ASA, Al 6061	3D-printed ASA; Al 6061 where loaded	Weather-stable, quick iteration
Mast	GI pipe, aluminium, FRP	GI 32 NB, medium class	Stiff, cheap, locally available
Radiation shield	White ASA louvres, aluminium	White ASA louvres	Passive cooling, low weight
Fasteners	Zinc plated, SS304	SS304 (A2)	Corrosion in monsoon and dew

The mast is the critical structure: the node is a small sail on a cantilever, so wind bending at the base governs the design.

Wind-load check (hand calc)

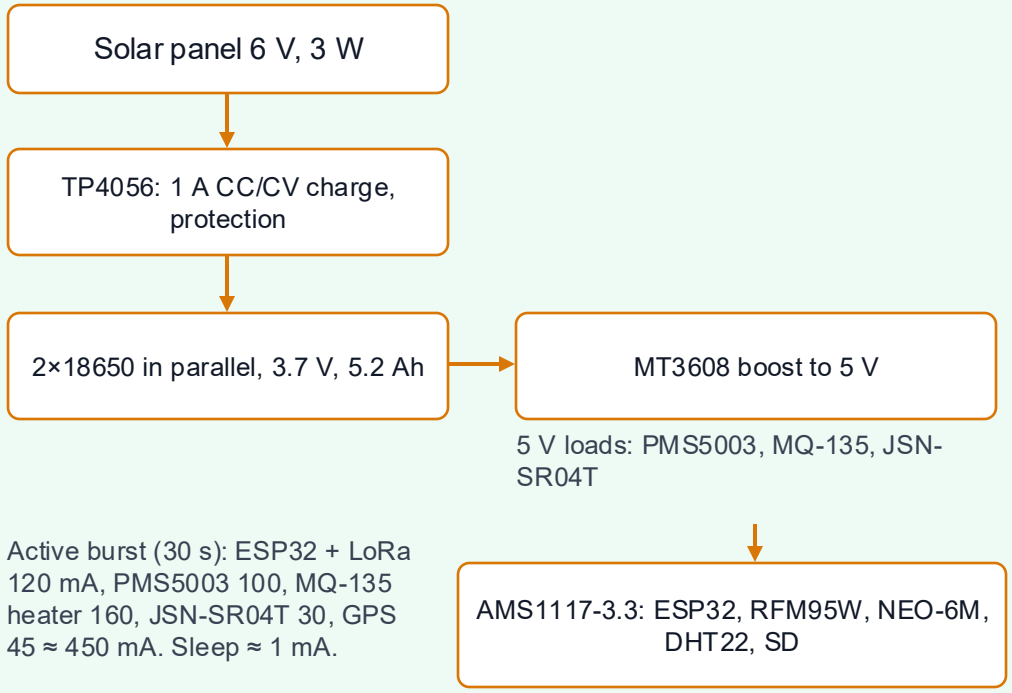
- Design wind 50 m/s: $q = \frac{1}{2}\rho v^2 = 1.5$ kPa
- Area ≈ 0.068 m² (box 0.032, panel 0.021, shield 0.015), Cd 1.2
- F ≈ 122 N at ~2 m, so M ≈ 245 N·m
- 32 NB pipe (OD 42.4, wall 3.2 mm): Z ≈ 3.59 cm³
- $\sigma = M/Z \approx 68$ MPa vs 210 MPa assumed yield, FoS ≈ 3.1

Next step: this is a screening calculation, not FEA. Static FEA with mesh-convergence plots on the CAD model (foot bracket and boom root) is planned; site wind zone to be confirmed per IS 875 Part 3.

Electronics: Pinout & Power Distribution

Signal	ESP32 GPIO	Note
DHT22 DATA	GPIO4	10 kΩ pull-up
PMS5003 TX / RX	GPIO16 / 17	UART2, 9600 baud
PMS5003 SET	GPIO25	Low = sleep
MQ-135 AO	GPIO34 (ADC1)	10 k / 20 k divider
MQ-135 heater gate	GPIO26	MOSFET switch
JSN-SR04T TRIG	GPIO21	3.3 V logic
JSN-SR04T ECHO	GPIO35	Divider from 5 V
NEO-6M TX / RX	GPIO32 / 33	UART1 remapped
RFM95W SCK/MISO/MOSI	GPIO18 / 19 / 23	VSPI, 3.3 V
RFM95W NSS/RST/DIO0	GPIO5 / 14 / 27	Interrupt on DIO0
microSD CS	GPIO13	Shared SPI bus
Battery sense	GPIO36 (ADC1)	100 k / 100 k divider

Power tree and voltage regulation



Design notes: 100–470 µF bulk capacitor near the LoRa module for transmit bursts; brownout detector and task watchdog enabled; MQ-135 heater and PMS5003 fan only powered during the sampling window.

Itemized Bill of Materials: Environmental Node

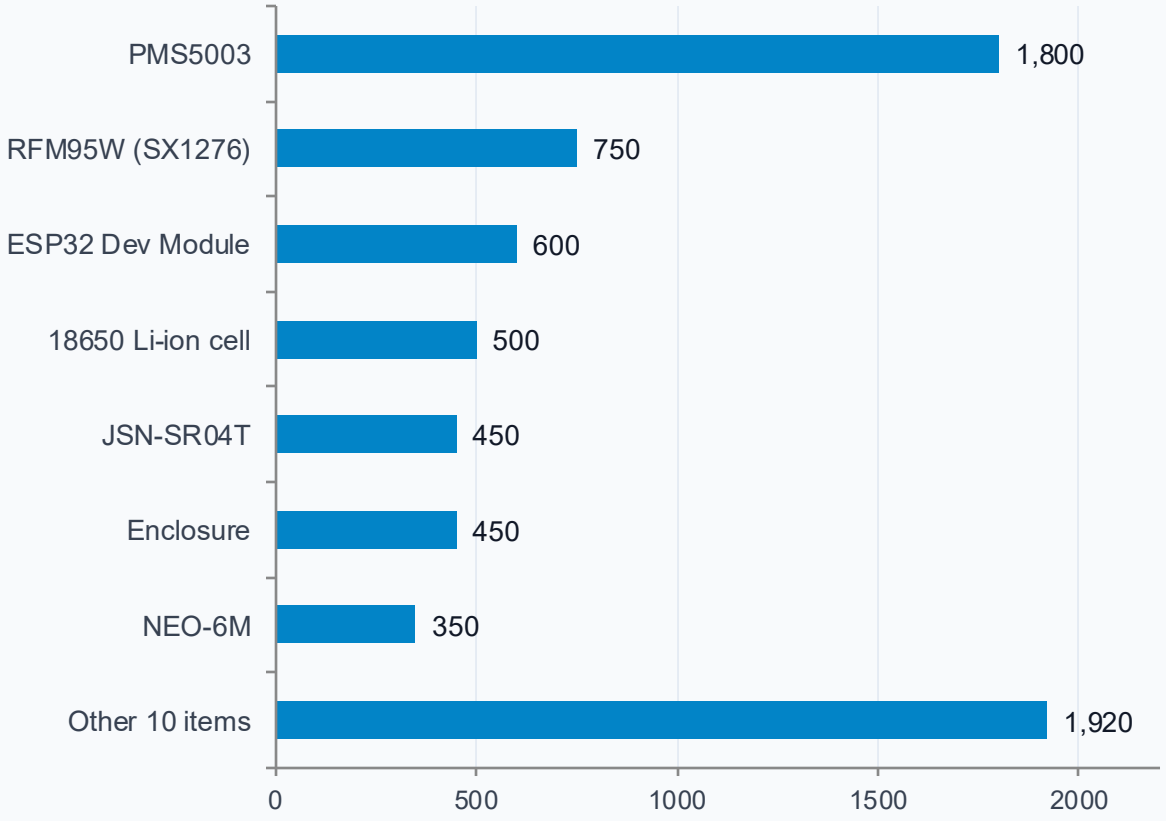
ELECTRONICS & BOM

#	Item	Specification	Qty	Unit (₹)	Total (₹)	Lead time	Source
1	ESP32 Dev Module	ESP32-WROOM-32	1	600	600	3–5 d	Robu.in (IN stock)
2	DHT22	Temperature & humidity	1	180	180	3–7 d	Amazon.in (IN stock)
3	MQ-135	Gas / air-quality index sensor	1	150	150	3–7 d	Robu.in (IN stock)
4	PMS5003	Laser PM2.5 sensor	1	1,800	1,800	7–10 d	ElectronicsHub (IN stock)
5	JSN-SR04T *	Waterproof ultrasonic, IP67 probe	1	450	450	3–7 d	Amazon.in (IN stock)
6	RFM95W (SX1276) *	LoRa module, set to 865–867 MHz	1	750	750	5–10 d	Robu.in / Amazon.in (IN)
7	NEO-6M	GPS module	1	350	350	5–7 d	Robu.in (IN stock)
8	MicroSD module + card	Local data storage	1	300	300	5–7 d	Amazon.in (IN stock)
9	TP4056	Li-ion charge module	1	50	50	2–5 d	Amazon.in (IN stock)
10	MT3608	Boost converter to 5 V	1	70	70	2–5 d	Amazon.in (IN stock)
11	AMS1117-3.3	3.3 V LDO regulator	1	20	20	2–5 d	Robu.in (IN stock)
12	Custom PCB	FR-4, 2-layer	1	300	300	7–10 d	JLCPCB (import)
13	18650 Li-ion cell *	3.7 V, 2600 mAh (2 in parallel)	2	250	500	3–7 d	Amazon.in (IN stock)
14	Solar panel *	6 V, 3 W	1	350	350	3–7 d	Amazon.in (IN stock)
15	Enclosure	IP65, ABS plastic	1	450	450	5–10 d	Local vendor
16	GI mast, 32 NB × 2 m *	Pole + clamps	1	350	350	3–7 d	Local vendor
17	Antenna, glands, fasteners *	SMA antenna, PG7, SS screws	1	150	150	3–7 d	Amazon.in (IN stock)
	Total estimated cost (one node)		**18**		**6,820**		

** Added or revised against BOM v1: RFM95W set to 865–867 MHz replaces the 433 MHz Ra-02; waterproof JSN-SR04T replaces HC-SR04; solar panel, mast and second cell added for field use. Starred prices are estimates; re-quote before ordering. Gateway is costed separately (slide 14).*

Procurement Sourcing & Cost Analysis

Cost by item (₹, per node)



Cluster economics (estimate)

- Node: ₹6,820; gateway (ESP32, LoRa, GSM, enclosure, pole): about ₹3,500
- 10 nodes + 1 gateway ≈ ₹71,700, about 1/139 of one ₹1 crore reference station

Sourcing flags

- **Local stock:** all modules and cells from Indian resellers (Robu.in, Amazon.in, ElectronicsHub), 2–10 days.
- **Imported:** PCB from JLCPCB, 7–10 days plus shipping; order first.
- **Largest risk:** PMS5003 is 26% of cost and has the longest module lead time (7–10 days).
- **Plan:** order one spare of each sensor; RFM95W must be the 865–867 MHz-capable variant.

Cluster figures are order-of-magnitude estimates from BOM v1 prices and new quotes still to be confirmed; reference-station cost from The Wire Science.

Your Maker Journey: Evidence & Repository

Authenticity verification

Insert photo: team members modelling, soldering and assembling, ID badges visible

- High-resolution lab photos of team members at work
- Institutional ID badges in frame
- Joint lab review photo with faculty mentor, if attached

Work-in-progress software evidence

Insert screenshot: CAD feature tree, KiCad DRC log or FEA mesh plot

- CAD feature tree with unsuppressed sketches
- KiCad layout with DRC rule-check log
- FEA convergence plots; TinyML training curve and confusion matrix

Raw repository linkage

Insert screenshot or QR: Drive / GitHub repository

- Link: add Google Drive or GitHub URL here
- Native files: .STEP, .SLDPRT, .kicad_sch, firmware, .tflite, BOM.xlsx
- Timestamps audited; external links only, no embedded video

Design consolidation (v1 documents → v2 submission): Ra-02 (433 MHz class) → RFM95W on 865–867 MHz; HC-SR04 → waterproof JSN-SR04T; IR flame sensor, MQ-7 and OLED dropped in favour of PMS5003 plus multi-sensor fusion; BOM re-costed.

Safety & Hazard Mitigation

Design FMEA (failure mode and effects analysis)

Failure mode	Hazard	Embedded fail-safe / mitigation
Li-ion over-charge, deep discharge or short	Cell heating; ignition risk in a forest	Protected 18650 cells, TP4056 protection, 2 A inline fuse, firmware cut-off at 3.0 V
Sensor drift or failure	False or missed alarm	24–48 h burn-in, plausibility checks, last-valid fallback, sensor-fault flag
LoRa link or gateway loss	Alert not delivered	ACK with retry and back-off, SD buffer, relay node, “node silent” alert on missed heartbeat
Brownout or MCU hang	Reset mid-write, lost alert	Brownout detector, task watchdog, bulk capacitor at LoRa transmit
Water ingress, condensation	Corrosion, short circuit	IP65 enclosure, gasketed PG7 glands, conformal coat, desiccant

Standards compliance

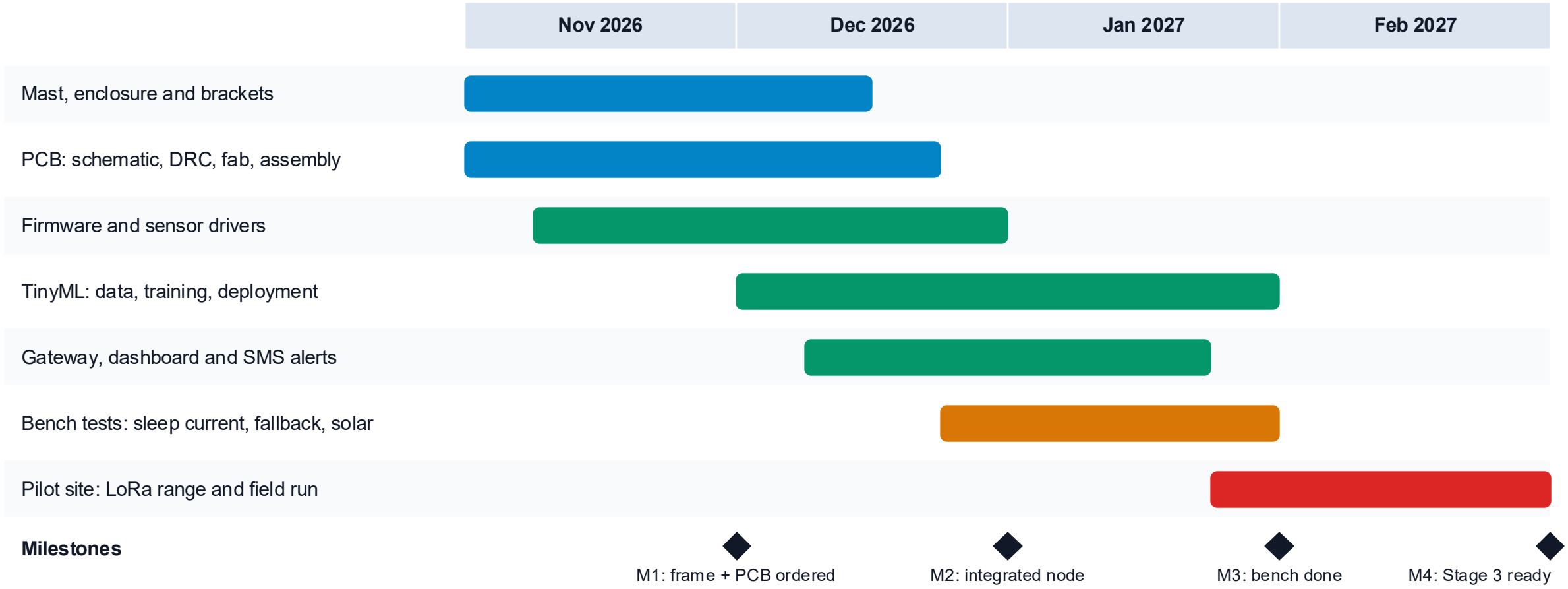
- **IP65:** IEC 60529 ingress protection for the enclosure.
- **Li-ion cells:** IEC 62133-2 (BIS IS 16046 Part 2) certified cells only.
- **Radio:** WPC/DoT 865–867 MHz rules (GSR 564(E)): ≤ 1 W, 4 W ERP, 200 kHz; LoRaWAN IN865 plan.
- **Air quality:** CPCB NAAQS and National AQI bands for reporting.
- **EMC / ESD:** IEC 61000-4-2 as a design guide.

Considered in design; no formal certification is claimed yet.

Hazard note: the node sits in forests and floodplains, so cell safety and enclosure sealing are treated as the highest-severity items.

Roadmap: 4-Month Fabrication Plan

ROADMAP



Stage 3 target: functional 3-node + gateway prototype with logged bench results, ready for live field testing at a pilot site by the end of Month 4. Month labels assume work starts November 2026; adjust to the confirmed schedule.

Technical Challenges & Risks

CHALLENGES & RISKS

Bottleneck: LoRa range in forest

Observed risk: Dense foliage can cut range from an estimated 2–5 km in the open to roughly 0.5–1 km, leaving remote nodes out of gateway reach.

Impact: coverage holes exactly where fires start. Link budget: 17 dBm + 7 dBi antennas vs −132 dBm sensitivity (SF10) gives 156 dB, or 146 dB after a 10 dB fade margin.

Bottleneck: Battery drain

Observed risk: All sensors running continuously draw ~700–800 mA, so a 2000 mAh cell would last only about 2.5–3 hours without sleep scheduling.

Impact: nodes would die within hours instead of the ~4.5-day target, which is unacceptable at unattended sites.

Bottleneck: False alarms and drift

Observed risk: MQ-135 readings shift with humidity, temperature and ageing; single-sensor thresholds can trigger false fire or pollution alerts.

Impact: alert fatigue; a village that gets false alarms stops trusting the system.

Bottleneck: Heat and humidity bias

Observed risk: Sun-heated enclosures bias DHT22 readings; PMS5003 over-reads at high humidity; ultrasonic range varies ~3.5% between 20 and 40 °C.

Impact: uncorrected, a 100 cm water-level reading can be off by about 3.5 cm.

Note: all range and runtime values are design-stage estimates from datasheets and calculations; field verification is planned for Stage 3.

Potential Solutions & Risk Mitigation

Mitigation: LoRa range

Gateway on a 10–15 m pole, 500 m–1 km node spacing, relay-node provision, and SF10–SF12 for longer reach (alerts are short messages). Fallback: SX1262 module for extra link margin.

Pilot target: ≥ 95% packet delivery at the planned spacing.

Mitigation: Battery life

Deep-sleep duty cycling with sensors powered only while reading, solar charging and a 2×18650 pack; battery voltage reported in every packet.

Pilot target: ≥ 7 days autonomy with solar over the field week.

Mitigation: False alarms

Multi-sensor fusion (alert only when temperature rise and smoke or gas agree) plus Edge AI classification instead of fixed thresholds.

Pilot target: ≤ 1 false alarm per node per month, recall ≥ 90%.

Mitigation: Drift and heat

24–48 h sensor burn-in, periodic recalibration, ventilated radiation shield, RH correction for PM2.5, speed-of-sound correction for the level sensor, and fallback to last valid reading. Alternates: SHT31, SPS30.

Pilot target: PM2.5 within ±20% of a collocated reference after calibration.

Verification plan: bench tests (sleep current, sensor fallback) and a pilot-site LoRa range test are planned for Stage 3; no results are claimed yet.

References & Sources

REFERENCES

[1] NDMA flood management guidelines (summary): vikaspedia.in/social-welfare/disaster-management-1/natural-disasters/floods

[2] Down To Earth, “It’s been 40 years! Update India’s flood map NOW”: downtoearth.org.in

[3] Rajya Sabha PQ 267, Forest Fires, MoEFCC, 13 Mar 2025 (ISFR 2023: 27.97%): rsdebate.nic.in

[4] IndiaSpend, “Rising forest fires could hinder India’s green cover ambitions” (34,562 km² burnt; ~9 in 10 fires human-caused): indiaspend.com

[5] Tribune India, FSI fire-season report (11.34% of forest and scrub in extreme/very-high zone): tribuneindia.com

[6] Krishna B. et al., Lancet Planetary Health 2024, doi:10.1016/S2542-5196(24)00114-1

[7] Mongabay-India, Jul 2024, NAAQS 60 vs WHO 15 µg/m³: india.mongabay.com

[8] Eos 103 (2022), “Indian cities invest in low-cost air quality sensors”, doi:10.1029/2022EO220495

[9] The Wire Science, cost of a continuous air-monitoring station (~₹1 crore + 10%): science.thewire.in

[10] DoT / WPC Wing clarification on 865–867 MHz equipment (GSR 564(E)), 27 Mar 2019: dot.gov.in

[11] LoRa Alliance, RP002 Regional Parameters (IN865 plan); Semtech SX1276 datasheet

[12] Datasheets: Espressif ESP32; Plantower PMS5003; Aosong DHT22; u-blox NEO-6; JSN-SR04T

[13] Standards: IEC 60529; IEC 62133-2; IS 875 Part 3; CPCB NAAQS 2009 and National AQI

Image sources: no third-party images are used; all diagrams, charts and concept views were created by the team for this submission.